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Foodborne outbreaks of norovirus in England, 2015 to 2023

It is well documented that every year norovirus outbreaks, driven by person-to-person transmission, contribute to winter pressures in health and social care settings (1). However, there is less awareness of the burden associated with foodborne norovirus outbreaks. Foodborne transmission of norovirus through the consumption of contaminated food, either during the production process or preparation, has been estimated to account for around 12% of annual norovirus cases in the UK (2). This article summarises foodborne outbreaks of norovirus which were reported to the UK Health and Security Agency's (UKHSA) Electronic Foodborne and non-foodborne Outbreak Surveillance System (EFOSS) in England between 2015 and 2023.

UKHSA is responsible for the detection, investigation and management of outbreaks of foodborne disease in England, working closely with Public Health Wales, Public Health Scotland and the Public Health Agency of Northern Ireland, as well as food safety, animal health and local authority colleagues to implement control measures to prevent further cases. A foodborne outbreak is defined as 'an incidence, observed under given circumstances, of 2 or more human cases of the same disease and/or infection, or a situation in which the observed number of human cases exceeds the expected number and where the cases are linked, or are probably linked, to the same food source (including potable water)' (Directive 2003/99/EC) (3). Data on foodborne outbreaks of norovirus between 2015 and 2023 was previously published in the 2021 and 2024 (4, 5) UK Food Security Reports. However, in these reports data was presented at the UK level and aggregated with outbreaks of other enteric viruses.

Trends in foodborne norovirus outbreaks

Overall, 67 foodborne norovirus outbreaks in England were reported to eFOSS between 2015 and 2023 (Figure 1). Of the 4,018 cases associated with these outbreaks, 7.4% (n=299) were laboratory confirmed as norovirus and less than 1% (n=11) of cases were hospitalised. The number of cases per outbreak ranged from 3 to 1,000 with a median of 31 cases per outbreak, however this varied across the years (Table 1). 2020 marked the start of the COVID-19 pandemic with only 3 outbreaks reported that year, a notable decrease compared to 2019 during which 15 foodborne norovirus outbreaks were reported. It is likely the COVID-19 pandemic restrictions, particularly those affecting the hospitality and catering sector, contributed to a drop in reported outbreaks during 2020 and into 2021.

Figure 1: Number of reported foodborne norovirus outbreaks by vehicle category and year

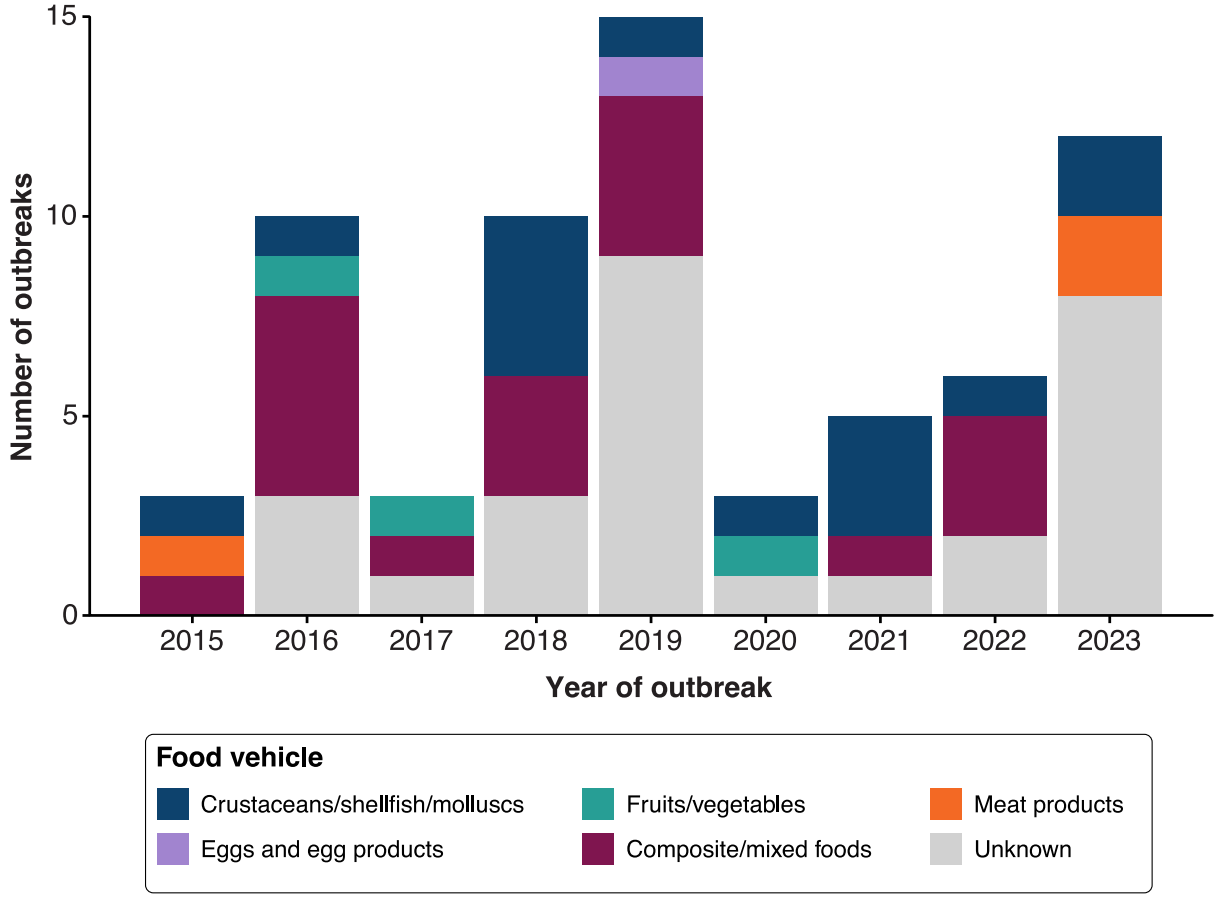

Table 1. Number of reported foodborne norovirus outbreaks, cases, laboratory confirmed cases and hospital admissions, by year

Year	Number of outbreaks	Number of affected cases	Number (%) of laboratory-confirmed cases	Number (%) of cases admitted to hospital	Median (range) affected cases per outbreak
2015	3	210	23 (11.0%)	0 (0.0%)	73 (17 – 120)
2016	10	1,407	80 (5.7%)	2 (0.1%)	35 (20 – 1000)
2017	3	150	32 (21.3%)	1 (0.7%)	51 (29 – 70)
2018	10	310	34 (11.0%)	0 (0.0%)	33.5 (3 – 50)
2019	15	457	54 (11.8%)	4 (0.9%)	22 (13 – 70)

Year	Number of outbreaks	Number of affected cases	Number (%) of laboratory-confirmed cases	Number (%) of cases admitted to hospital	Median (range) affected cases per outbreak
2020	3	235	15 (6.4%)	0 (0.0%)	55 (23 – 157)
2021	5	501	23 (4.6%)	1 (0.2%)	94 (20 – 209)
2022	6	261	10 (3.8%)	2 (0.8%)	28 (5 – 131)
2023	12	487	28 (5.7%)	1 (0.2%)	24 (4 – 184)
Total	**67**	**4,018**	**299 (7.4%)**	**11 (0.3%)**	**31 (3 – 1000)**

A food vehicle was implicated in 58.2% (n=39) of the investigated outbreaks. The most commonly reported food categories were composite/mixed foods (18 outbreaks, 26.9%) and crustaceans/shellfish/molluscs (14 outbreaks, 20.9%). Oysters were the most frequently implicated single food item, associated with 1,117 cases across 13 outbreaks (Figure 1, Table 2).

Table 2: Number of foodborne norovirus reported outbreaks and affected cases, by vehicle category and year

Year	Crustaceans/ shellfish/ molluscs: outbreaks (cases)	Meat products: outbreaks (cases)	Fruits/ vegetables: outbreaks (cases)	Eggs and egg products: outbreaks (cases)	Composit mixe foods outbreak (case
2015	1 (17)	0 (0)	1 (73)	0 (0)	1 (12
2016	1 (100)	1 (23)	0 (0)	0 (0)	5 (3
2017	0 (0)	1 (70)	0 (0)	0 (0)	1 (2
2018	4 (115)	0 (0)	0 (0)	0 (0)	3 (10
2019	1 (58)	0 (0)	0 (0)	1 (30)	4 (8

Year	Crustaceans/ shellfish/ molluscs: outbreaks (cases)	Meat products: outbreaks (cases)	Fruits/ vegetables: outbreaks (cases)	Eggs and egg products: outbreaks (cases)	Composit mixe foods outbreak (case
2020	1 (157)	1 (55)	0 (0)	0 (0)	0 (
2021	3 (449)	0 (0)	0 (0)	0 (0)	1 (2
2022	1 (65)	0 (0)	0 (0)	0 (0)	3 (15
2023	2 (188)	0 (0)	2 (81)	0 (0)	0 (
Total	**14 (1,149)**	**3 (148)**	**3 (154)**	**1 (30)**	**18 (64**

83.6% (n=56) of reported outbreaks were associated with catering settings (restaurants/food service establishments, takeaways or fast-food outlets), accounting for 52.1% (n=2,093) of the total affected cases. While the majority of reported outbreaks were linked to a specific food service establishment, for 6 of the largest reported outbreaks, the setting was designated as ‘multiple places of exposure’, therefore indicating that a contaminated food product that caused the outbreak was consumed at multiple locations such as multiple different food service establishments, institutions or in the home. For the largest outbreak, reported in 2016, it was not possible to identify the likely food vehicle due to the complex nature of the investigation, despite at least 1,000 cases being recorded (6). The 5 other outbreaks recorded as ‘multiple places of exposure’ were all linked to oysters between 2019 and 2022, ranging in size from 65 to 209 cases and accounting for 16.7% (n= 671) of total affected cases.

Table 3: Number of reported foodborne norovirus outbreaks and affected cases by setting and year

Year	Restaurant/ café/ pub/ bar/ hotel/ catering service: outbreaks (cases)	Institutional/ residential: outbreaks (cases)	School or kindergarten: outbreaks (cases)	Multiple places of exposure: outbreaks (cases)	Ot foodbo setti outbrea (cas
2015	2 (193)	0 (0)	0 (0)	0 (0)	1 (
2016	9 (407)	0 (0)	0 (0)	1 (1)	0
2017	3 (150)	0 (0)	0 (0)	0 (0)	0

Year	Restaurant/ café/ pub/ bar/ hotel/ catering service: outbreaks (cases)	Institutional/ residential: outbreaks (cases)	School or kindergarten: outbreaks (cases)	Multiple places of exposure: outbreaks (cases)	Ot foodbo setti outbre (cas
2018	10 (310)	0 (0)	0 (0)	0 (0)	0
2019	13 (371)	0 (0)	10 (56)	0 (0)	1 (
2020	2 (78)	0 (0)	0 (0)	1 (157)	0
2021	1 (32)	10 (20)	0 (0)	3 (449)	0
2022	4 (65)	10 (131)	0 (0)	1 (65)	0
2023	12 (487)	0 (0)	0 (0)	0 (0)	0
Total	**56 (2,093)**	**2 (151)**	**1 (56)**	**6 (1,671)**	**2 (**

Outbreak investigations also aim to gain insight into other factors which led to the cases of illness to inform risk management and prevention strategies. Involvement of an infected food handler, use of an unprocessed contaminated ingredient and cross-contamination were the most frequently reported contributing factors, highlighting the importance of raising awareness with the public about the risks of foodborne norovirus and how to safely prepare and store food.

Food Standards Agency guidance on how to reduce your risk of food poisoning is available at food safety and hygiene guidance (https://www.food.gov.uk/food-safety-and-hygiene).

UKHSA guidance on how to prevent onward norovirus transmission is available at: Norovirus: What to do if you catch it and helping to stop the spread (https://ukhsa.blog.gov.uk/2022/11/17/norovirus-what-to-do-if-you-catch-it-and-helping-to-stop-the-spread/).

Underreporting of cases and foodborne outbreaks is a known challenge in the surveillance of gastrointestinal infections. During a foodborne outbreak investigation, UKHSA works with partner agencies to collate epidemiological, food chain and microbiological evidence to identify the likely food vehicle or source of contamination and the level of evidence obtained can vary notably between outbreaks. When interpreting the above data the following limitations should be considered:

- not all individuals linked to outbreaks will have laboratory confirmation of illness
- hospital admission status is only known for cases who received public health follow up or were contacted as part of the investigation
- not all outbreaks of foodborne illness will be investigated and it is not always possible to identify the likely vehicle or source of contamination
- not all outbreaks are microbiologically linked to the implicated food vehicle through laboratory testing - suspected vehicles may be reported when the implicated food was not available to test in laboratory
- outbreaks reported in England may include cases who are resident in other parts of the UK or outside the UK
- EFOSS is a live reporting system and outbreaks are only reported once the investigation has concluded, as such data for previous years may be updated retrospectively and the number of outbreaks may rise compared to previous outputs

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Infection reports in this issue

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Laboratory-confirmed cases of invasive meningococcal infection in England: April to June 2025 (https://www.gov.uk/government/publications/meningococcal-disease-laboratory-confirmed-cases-in-england-2024-to-2025)

Campylobacter and non-typhoidal salmonella infections in England: Q1 2023 to Q2 2025 (https://www.gov.uk/government/publications/gastrointestinal-infections-in-england)

Carbapenemase-producing Gram-negative organisms in England since October 2020: quarterly update, Q2 2025 (https://www.gov.uk/government/publications/carbapenemase-producing-gram-negative-bacteria-laboratory-surveillance)

Vaccine coverage reports

RSV vaccine coverage report in older adults for catch-up cohorts in England: August 2025 (https://www.gov.uk/government/publications/rsv-older-adults-vaccination-coverage-in-england)

RSV maternal vaccination coverage in England: May 2025 (https://www.gov.uk/government/publications/rsv-maternal-vaccination-coverage-in-england)

Prenatal pertussis vaccination coverage in England from April to June 2025 (https://www.gov.uk/government/publications/pertussis-immunisation-in-pregnancy-vaccine-coverage-estimates-in-england-october-2013-to-march-2014)

Quarterly vaccination coverage statistics for children aged up to 5 years in the UK (COVER programme): April to June 2025

(https://www.gov.uk/government/statistics/cover-of-vaccination-evaluated-rapidly-cover-programme-2025-to-2026-quarterly-data)

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